CCH® Tagetik 5.3 Hardware and Software Architecture Overview

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CCH® Tagetik

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Introduction

This architecture overview presents a high level technical overview of the latest version of CCH® Tagetik regarding the hardware and software architecture and requirements of the latest released version and Service Pack of CCH Tagetik, 5.3.SP17, unless otherwise noted.

The intention of this document is to provide a technical overview of the essential points regarding the hardware and software infrastructure architecture of CCH Tagetik with a particular focus on on-premises and private cloud deployments of the application.

For details regarding specific compatibility with third party software, please see the server and the client compatibility matrices which are available on the CCH Tagetik Community web portal (https://community.tagetik.com/product/knowledge-base/architecture/tagetik/compatibility). These documents also contain the compatibility roadmap for future service packs.

For specific information regarding CCH Tagetik Cloud architecture, please see the Cloud Description documents available on the CCH Tagetik Community (https://community.tagetik.com/legal/cch-tagetik-cloud-specifications).

Pre-Packaged Solutions

Pre-Packaged Solutions, such as Account Reconciliation, PCPC, Lease Accounting (IFRS 16), IFRS 17, Solvency II, etc., may have different compatibilities with third party software. In particular, for IFRS 16 projects, please consult the addendum to this document named *CCH Tagetik Lease Accounting Solution Addendum.* All documents are available on the Community.

Please contact CCH Tagetik with any questions or comments related to the contents of this document.

Architecture Overview

CCH Tagetik is an enterprise application written in Java according to the JEE specifications and platform. JEE provides an environment for developing and running enterprise software, including network and web services, and other large-scale, multi-tiered, scalable, reliable, and secure network applications. The application is comprised of three tiers:

- An Application Server Tier requiring a JEE compliant Application Server.
- A RDBMS Tier requiring a compatible RDBMS system.
- A Client Tier which consists in three interfaces, a web GUI, .Net Microsoft Office Add-ins for Excel, Word and PowerPoint, and an API layer based on REST.

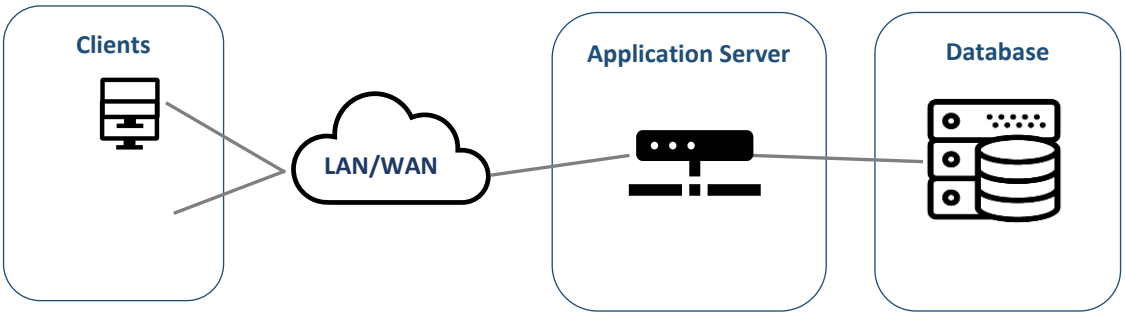


Figure 1 Standard Infrastructure Architecture for the Application.

Application Server and JDKs Compatibility

CCH Tagetik software requires an application server compliant with the JEE specification. To date compliance has been tested with the generally available JEE application servers and JDKs. To date, the application is compatible with the following JEE compatible AS.

- JBoss EAP
- WildFly
- Oracle WebLogic
- IBM WebSphere

Specific details about compatible versions and compatible JDKs are available on the CCH Tagetik Community web portal.

Database Server Compatibility

CCH Tagetik requires a relational database to store all application data and configurations. The following vendor RDBMS's are supported for the current service pack.

- Oracle Database
- Microsoft SQL Server
- PostgreSQL
- SAP Hana

Details regarding the versions of each RDBMS are available in the compatibility matrices available on the CCH Tagetik Community web portal.

Operating System Compatibility

Application Server

The CCH Tagetik application is not dependent on the operating system of the application server. As a result, it is portable among any operating system supported by the compatible application server utilized. To date portability has been verified with both Windows and Unix-like operating systems such as AIX, Linux, Solaris, etc.

Database Server

It is possible to use any of the OS's supported by the above-mentioned RDBMS. In general, the vendor standard edition or equivalent is adequate. However, due to a customer's project specifications and infrastructure requirements, a different edition may be necessary. If there are any questions, please consult with the project manager or CCH Tagetik. As mentioned in the introduction, pre-packaged solutions may have different compatibilities. Please the relevant documentation for those solutions.

Client PC

The application can be accessed by one of two user interfaces. The web user interface can be utilized with any compatible browser on any operating system. The .Net Add-ins user interface requires Office for Windows.

Application Installation and Deployment

The application is provided as a JEE EAR archive file which must be deployed on compatible application server(s). SQL scripts are utilized to create the necessary objects in the selected RDBMS. Both the ear archive and the SQL scripts are available from CCH Tagetik or from the CCH Tagetik Community web portal.

Application Initialization

CCH Tagetik is contained in a single web application and is initialized during the Servlet Container startup process. The initialization process is started by means of a ServletContextListener suitably configured in the web.xml file. During the initialization process some JNDI resources (e.g., for the database connections) are created. The AS requires JDBC compatible drivers to communication with the databases. The server compatibility matrix includes a table of compatible drivers.

Analytics (for Microsoft Analysis Services)

CCH Tagetik Analytics requires an instance of SQL Server Analysis Services. In addition, a certain edition may be required based on the requirements of the project and the characteristics of the infrastructure. The following comparison outlines the differences between SQL Server Editions.

SSAS Standard Edition

- Max 128 GB RAM, max 4 sockets/24 cores
- Allows partitions to be processed sequentially.
- Allows only the Scenario and Period CCH Tagetik models to be deployed, not the Time Version ones.
- Does not have Tabular models.

SSAS Enterprise Edition

- Has all the features of the Standard Edition, but
- Does not have limitations in number of cores and RAM,
- Allows parallel processing of multiple partitions,
- Allows tabular models,
- Allows Time Version models to be used.

Data Source Connections

Connections to the primary application data sources are made with JDBC objects and are obtained through JNDI lookup. All connection parameters including each database's access credentials are configured at the application server tier.

Integration with External Data Sources

CCH Tagetik provides multiple built-in tools for integrating with data sources. These tools include:

- ETL: The highly configurable ETL engine which permits data extraction, transformation and loading from relational database connections, files (i.e. text, Excel, XML), WebDAV, OData, and SFTP or HTTP/HTTPS endpoints as sources. It is also possible to write data to all of these channels.
- QDL: The Quick Data Loader permits quick access to data in files providing a drag and drop approach to data integration.

Environment Migration

CCH Tagetik provides built-in tools for migrating data, configurations, and metadata between environments.

- MDM: The Master Data Management permits import and export of project metadata.
- Import/Export: The built-in Import/Export tool allows proprietary imports and exports of table data.

Administration, Monitoring and Logging Tools

The application includes a built-in audit tool and log accessible via the web user interface. The audit tool tracks all application and user activity. All audit information is stored in the RDBMS tier.

Standard tools for both JEE application and relational database monitoring are supported. On the application server tier the application performs logging by means of the native AS and "log4j2" logging frameworks. It is possible to fully configure the logging levels as well as the logging devices (including mail mechanisms, network logging and so on). If no third party monitoring tool is available, it is possible to employ those provided with the "log4j" framework.

Backups

All application and business data is stored in databases on the database server. Any standard database backup utility or tool can be used, as long as it is compatible with the RDBMS. More details regarding backups and backup policies can be found in the CCH Tagetik Maintenance manual on the CCH Tagetik Community web portal.

High Availability Architecture

It is possible to implement a high availability architecture involving all CCH Tagetik server components, according to the JEE Application Server and the RDBMS architectures. Such a configuration expects several application servers to rely on the same database server, which in turn could also be a cluster.

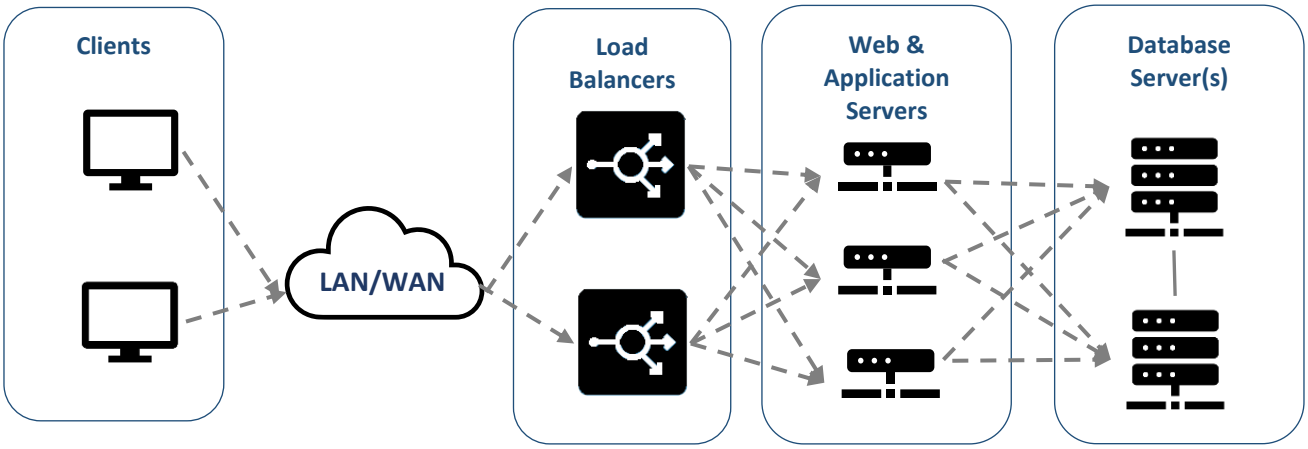


Figure 2 An example architecture of the infrastructure of the CCH Tagetik application with optional load balancer and optional clustered application and database server platforms.

If a load-balancer is deployed, it must implement “session affinity”. Any AS specific clustering mechanism is permitted only if it implements this.

Client PC Requirements

Requirements for the end-user PC should be interpreted on a per user basis where multiple users utilize the same workstation. A minimum of 2 GB of free disk space is required for prerequisites and other generic purposes. A multiple core processor and a minimum of 8 GB of RAM is suggested.

Display Resolution

The minimum recommended display resolution is 1280x1024. However, it is possible that other resolutions may also provide an acceptable user experience, such as 1920x1080, 1400x900, or 1280x800.

In addition, the utilization of very high-resolution displays (HD) often results in onscreen text which is difficult to read. Therefore, many users adjust font display rendering in Windows displays to increase readability. Unfortunately, CCH Tagetik .Net Add-ins are not compatible with Windows increased font rendering (display settings such as "125%" or "120DPI" and greater) as the increased font size causes layout issues with configuration windows preventing their proper usage. The workaround consists in setting the font display rendering back to 100% or "Normal".

Bandwidth

CCH Tagetik is compatible with recent versions of most web browsers available today. For the login page and the user interface HTML 5 is required. If exact versions are required, please contact CCH Tagetik Support.

Browser & OS	Version & Notes
MS Edge (Windows)	2 most recent major versions
Firefox	latest and extended support release ESR
Google Chrome	latest
Opera	2 most recent major versions
Safari	2 most recent major versions

Communication with the server is over standard HTTP/HTTPS protocols.

Bandwidth

Following the first connection to the web interface the browser loads javascript libraries from the server, the size of which is roughly 2 to 3 MB. This part of the application then manages all the subsequent HTTP/HTTPS user requests, aimed at the data upload/download. Concerning these requests (i.e. performed after the first connection), a minimum of 128Kbps is suggested for each user's connection. It is worth noting, however, that the lack of bandwidth is likely to result in a diminished user experience due to the increased response times.

CCH Tagetik Add-ins

In addition to the web browser interface, CCH Tagetik provides .Net Add-ins for Microsoft Office components such as Excel, Word, and PowerPoint. The .Net Add-ins have the some requirements.

- Microsoft Office 32/64 bit for Windows

Furthermore, the .NET Add-ins require the following components to be installed on the end user's workstation:

- Windows Installer
- .NET Framework
- Microsoft Visual Studio 2010 Tool for Office Runtime

Please see the CCH Tagetik Client Compatibility Matrix for all the details regarding the versions and requirements.

Example Application and Database Hardware Sizing

The hardware resource estimate described in this section is provided as an example of the resources that a small project might use with light concurrent usage, i.e. 5 users at a time. Actual projects might require very different resources depending on the number of users, the quantity and characteristics of the data to be processed and the output and data entry reports being utilized among other aspects. It is important to note that each project is different and specific recommendations regarding a particular project depend on factors not considered in the table below. CCH Tagetik can provide a sizing recommendation based on project characteristics.
Before beginning a project or production installation, we strongly recommend contacting CCH Tagetik or your partner for an individually prepared resource sizing recommendation based on the characteristics of the specific project.

Application Server	
Processors:	6 cores - In order to minimize single user response times, we strongly recommend a CPU model which maximizes single thread performance.
RAM	16-24 GB
Storage	50 GB depending on the retention policy for system logs.
Operating System	Any compatible 64 bit OS
Database Server	
Processors	6 cores - In order to minimize single user response times, we strongly recommend a CPU model which maximizes single thread performance.
RAM	16-24 GB
Storage	Minimum 100 GB/DB or schema, Additional storage will be necessary for the RDBMS system files. Storage solutions should be chosen and designed with I/O performance in mind. For example, RAID 10 or the usage of SSD.
Operating System	Windows/Linux 64 bits

Virtualization

The application infrastructure can be placed on virtualized resources of any of the widely available virtualization solutions. On the RDBMS tier, CCH Tagetik suggests considering a server architecture which favors high performance. The application has been also successfully utilized with container technology. It should be noted that CCH Tagetik does not provide support for container configurations nor provide any items related to containerization.

Glossary

The following table provides a short glossary for the terminology which are used in this manual.

AS	Application Server
DB	Database
RDBMS	Relational Database Management System, i.e. MS SQL Server, Oracle, PostgreSQL, or SAP S/4 Hana
JEE	Java Enterprise Edition
EAR	JEE Enterprise Archive, a file format used in JEE applications.
JBoss / WildFly	JBoss and its successor, WildFly, are freely available open source application servers. JBoss EAP is offered by Red Hat.
ETL	Extraction, Transformation and Loading
Core	Most CPU's today have multiple cores, internal processing units.
HTML 5	CCH Tagetik web interface is being migrated to HTML5, the latest version of HTML.
SSD	Solid State Disks - a storage device containing non-volatile flash memory, used in place of a hard disk because of its much greater speed.

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